Meet the Devices

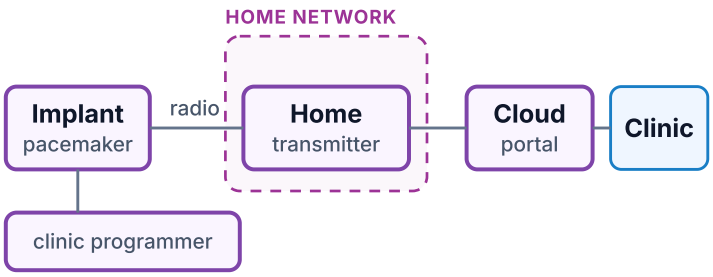
Four **fictional** products appear throughout the course. Each is a different kind of medical device with a different set of security challenges. Names, companies and details are invented; any resemblance to real products is coincidental.

In the sketches, ▭ marks parts the manufacturer builds and ⬚ dashed lines mark places where the device depends on something the manufacturer doesn't control.

IMPLANTABLE · HOME USE

CardiaPace

An implantable pacemaker, a bedside home transmitter, and a cloud remote-monitoring portal used by cardiology clinics.



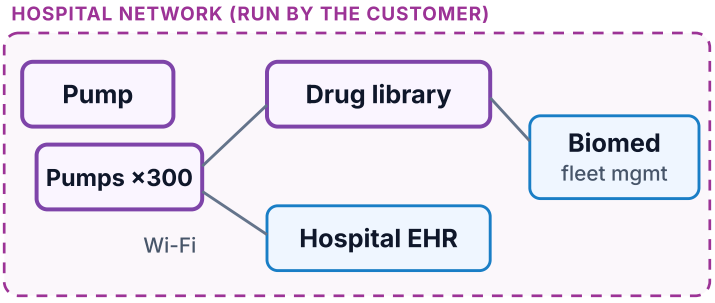
WHY SECURITY MATTERS HERE

- Life-sustaining therapy: integrity and availability come first.
- Implant firmware is hard to update, and patients can't simply switch it off.
- The transmitter lives on a home network the manufacturer doesn't control.
- One cloud portal holds data for many patients across many clinics.

HOSPITAL CONNECTED DEVICE

FlowGuard

A fleet of large-volume infusion pumps on hospital Wi-Fi, a drug-library server, and integration with the hospital's electronic health record (EHR) for auto-programming.



WHY SECURITY MATTERS HERE

- Hundreds of units on a network the hospital runs, not the manufacturer.
- A shared drug library: one bad update can affect every pump at once.
- Hospital security teams ask hard questions before they buy (questionnaires, the MDS2 form).
- Downtime means nurses fall back to manual programming.

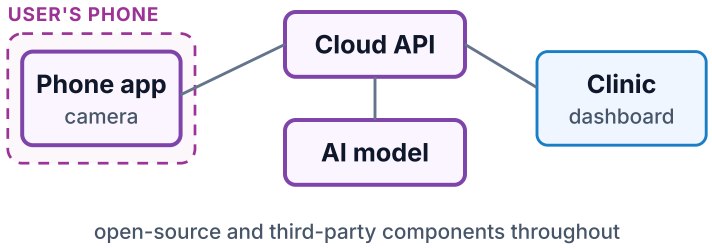
Using the cast

When you meet a new idea, ask how it would look for each device. The same principle, such as “design for updates”, plays out very differently for an implant, a hospital fleet, a phone app and a 15-year capital system. Then ask the same question of your own product.

SOFTWARE AS A MEDICAL DEVICE (SAMD)

DermaLens

An AI-assisted skin-lesion triage app: a phone camera app and a cloud model that suggests whether a lesion needs urgent dermatology review. Sold to clinics and, in some markets, used by patients.



open-source and third-party components throughout

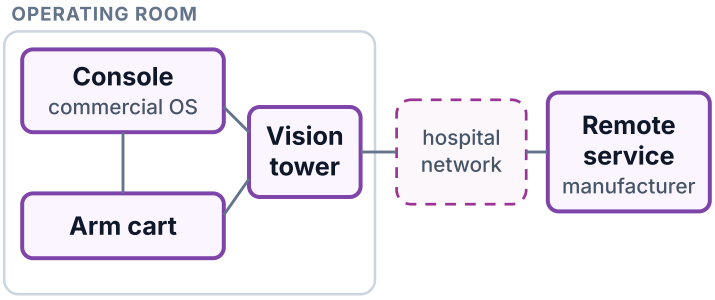
WHY SECURITY MATTERS HERE

- Runs on phones and cloud services the manufacturer doesn't fully control.
- Many open-source and third-party components, so the SBOM matters.
- Frequent releases: security has to keep pace with development.
- A tampered model or image pipeline gives wrong triage advice.

SURGICAL · CAPITAL EQUIPMENT

OrionArm

A surgical robotic system: a surgeon console running a commercial operating system, a patient-side arm cart, a vision tower, and remote service access for diagnostics and updates. Expected service life about 15 years.



WHY SECURITY MATTERS HERE

- A 15-year life outlasts operating-system and third-party support: legacy risk.
- Remote service access is powerful, so it needs strong control and logging.
- Downtime cancels surgeries: availability is a clinical issue.
- Updates need validation and scheduling with the hospital.

The four at a glance

DEVICE	ENVIRONMENT IT DEPENDS ON	HOW A FIX REACHES PATIENTS	WATCH MOST CLOSELY
CardiaPace	The patient's body and home network; the manufacturer's cloud	The transmitter updates itself; the implant usually needs a clinic visit	Integrity and availability of life-sustaining therapy
FlowGuard	A hospital-run network shared with hundreds of other devices	Scheduled with each hospital's biomedical engineers	The shared drug library, which reaches every pump
DermaLens	Users' phones and third-party cloud services	An app update can reach users within days	Third-party components; the integrity of the model and images
OrionArm	The operating room, the hospital network and remote service	Planned around surgery lists	Remote access; an operating system supported for less time than the robot

CardiaPace, FlowGuard, DermaLens and OrionArm are fictional devices created for training. Training material, not legal or regulatory advice.